\

METRIC
MIL-DTL-192C
01 August 2023
SUPERSEDING
MIL-A-192B(AR)
02 September 1965

MILITARY SPECIFICATION

AMMONIUM PERCHLORATE, TECHNICAL

MIL-DTL-192, dated 01 August 2023, is hereby reactivated and may be used for acquisition.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements and quality assurance provisions for the manufacture and acceptance of four grades and eight classes of ammonium perchlorate (NH_4ClO_4) (see 6.1).

Grade A – Low moisture content.
Grade B – Extra low moisture content.
Grade C – Tricalcium phosphate conditioner added.
Grade D – Tricalcium phosphate conditioner added.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must

Comments, suggestions, or questions on this document should be addressed to: U.S. Army DEVCOM Armaments Center, ATTN: FCDD-ACE-QSA, Picatinny Arsenal, New Jersey 07806-5000 or e-mailed to usarmy.picatinny.devcom-ac.list.ardec-stdzn-branch@army.mil. Since contact information can change, you may want to verify the currency of this information using ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 6810

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T05:28Z
Check the source to verify that this is the current version before use.

meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-286	-	Propellants, Solid: Sampling, Examination and Testing
MIL-STD-1916	-	DOD Preferred Methods for Acceptance of Product

(Copies of these documents are available online at https://quicksearch.dla.mil/.)

2.3 Non-Government standards and publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM D1193	-	Standard Specification for Reagent Water
ASTM E11	-	Standard Specification for Woven Wire Test Sieve Cloth and Test Sieves
ASTM E70	-	Standard Test Method for pH of Aqueous Solutions with the Glass Electrode
ASTM E300	-	Standard Practice for Sampling Industrial Chemicals.
ASTM E1064	-	Standard Test Method for Water in Organic Liquids by Coulometric Karl Fischer Titration

(Copies of these documents are available online at https://www.astm.org/.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Required inspections.

3.1.1 First article. When specified (see paragraph 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T05:28Z
Check the source to verify that this is the current version before use.

3.1.2 Conformance. A sample shall be subject to conformance inspection in accordance with 4.3.

3.2 Chemical requirements. The ammonium perchlorate shall conform to the applicable chemical characteristics as listed in Table I.

TABLE I. Chemical requirements for ammonium perchlorate.

Property	Grade A	Grade B	Grade C	Grade D
Total moisture, weight % (wt. %) maximum (max.)	0.08	0.05	0.08	0.10
Surface moisture, wt. % max.	0.020	0.015	0.020	----
Water insolubles, wt. % max.	0.03	0.01	0.25	0.20
Ash, sulfated, wt. % max.	0.25	0.15	0.45	----
Chlorate, as NH_4ClO_3, wt. % max.	0.02	0.02	0.02	0.02
Chloride, as NH_4Cl, wt. % max.	0.15	0.10	0.15	----
Sodium and potassium, wt. % max.	0.08	0.05	0.08	----
Calcium (3/) As elemental Ca As tricalcium phosphate, wt.% max.	 ---- ----	 ---- ----	 0.058 – 0.085 0.15 – 0.22	 0.058 – 0.085 0.15 – 0.22
Iron (2/, 3/) As elemental Fe As Fe_2O_3, wt. % max.	 0.0025 0.0036	 0.0025 0.0036	 0.0025 0.0036	 ---- ----
Assay, NH_4ClO_4, minimum (min.)	99.0	99.0	98.8	98.8
pH	4.3 – 5.3	4.3 – 5.3	5.5 – 6.5	5.0 – 6.5
Notes: 1/ "-----" indicates that there is no requirement. 2/ when required (see 6.2.f). 3/ Calcium and iron impurities may be measured in its elemental or compound form by the appropriate verification method.				

3.3 Granulation. Ammonium perchlorate shall conform to the granulation requirements as listed in Table II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T05:28Z
Check the source to verify that this is the current version before use.

TABLE II. Granulation.

U.S. Alternative Sieve Number (% thru) 1/	Class 1	Class 2	Class 3	Class 4	Class 5	Class 6	Class 7	Class 8
12	----	----	----	----	----	----	99.5 min.	----
18	----	----	----	99.5 min.	----	----	-	----
30	----	----	----	----	----	----	95.0 min.	----
40	99.5 min.	----	----	----	----	----	45.0 – 65.0	----
45	-	----	----	----	----	----	----	98.0 min.
50	85.0 – 98.0	93.0 min.	----	88.0 – 96.0	99.0 min.	89.0 – 97.0	3.0 max.	----
70	----	----	----	50.0 – 70.0	----	----	----	----
100	----	----	70 min.	17.0 – 38.0	----	18.0 – 50.0	----	----
140	----	----	----	2.0 – 15.0	1.0 min.	2.0 – 15.0	----	----
200	1.0 – 9.0	----	----	1.0 – 5.0	----	----	----	1.0 – 9.0
325	----	----	----	2.0 max.	----	----	----	8.0 max.

Notes:
1/ U.S. Alternative Sieves shall be in accordance with ASTM E11.
2/ "---- " indicates no requirement.

3.4 Workmanship. Ammonium perchlorate shall be an odorless white solid, uniform in quality, and free from visible impurities and foreign material such as dirt, wooden particles or any other foreign matter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T05:28Z
Check the source to verify that this is the current version before use.

4. VERIFICATION

TABLE III. Requirements/verification cross reference matrix.

Method of verification 1 – Analysis 2 – Demonstration		3 – Examination 4 – Test				Classes of verification A – First article B – Conformance		
Section 3 Requirement	Description	Verification Methods				Verification Class		Section 4 Verification
		1	2	3	4	A	B	
3.1.1	First article	X		X	X	X		4.2
3.1.2	Conformance	X		X	X		X	4.3
3.2	Total moisture	X			X	X	X	4.4.1.1
3.2	Surface moisture	X			X	X	X	4.4.1.2
3.2	Water insolubles	X			X	X	X	4.4.2
3.2	Ash, sulfated	X			X	X	X	4.4.3
3.3	Chlorate	X			X	X	X	4.4.4 or 4.4.5
3.2	Chloride	X			X	X	X	4.4.4 or 4.4.6
3.2	Sodium and potassium	X			X	X	X	4.4.7 or 4.4.8
3.2	Tricalcium phosphate	X			X	X	X	4.4.7 or 4.4.9
3.2	Iron	X			X	X	X	4.4.7 or 4.4.10
3.2	Assay	X			X	X	X	4.4.11
3.2	pH	X			X	X	X	4.4.12
3.3	Granulation			X		X	X	4.4.13
3.4	Workmanship			X		X	X	4.4.14

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see paragraph 4.2).
b. Conformance inspection (see paragraph 4.3).

4.2 First article inspection. When specified (see paragraph 6.2), a sample shall be subjected to first article verification in accordance with 4.3.3. Samples of not less than nine hundred (900) grams of ammonium perchlorate collected in accordance with 4.3.2 shall be submitted for examination. Samples shall be produced using the same process procedures, material supply sources, production equipment, and production line as will be used in regular production.

4.2.1 Rejection. If any sample fails to comply with any of the applicable requirements, the first article shall be rejected.

4.3 Conformance inspection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T05:28Z
Check the source to verify that this is the current version before use.

4.3.1 Inspection lot formation. Inspection lots and batches shall comply with the MIL-STD-1916, paragraph "Formation and identification of lots or batches." Each batch shall consist of a quantity of ammonium perchlorate that has been subjected to the same chemical and physical mixing process intended to make the final product homogeneous. A lot shall consist of one or more acceptable batches of ammonium perchlorate produced by one manufacturer in accordance with one specification and revision under one continuous set of operating conditions.

4.3.2 Ammonium perchlorate sampling. Sampling for inspection shall be performed either at the supplier location where the ammonium perchlorate is packaged, or at the receiving facility where the material is unpacked for use in the next higher sub-assembly or formulation.

4.3.2.1 Batch sampling at the supplier. Samples shall be collected from the beginning, middle, and end of every batch in every lot in accordance with ASTM E300 for solids. A composite sample of not less than 500 grams shall be formed which is assumed to be representative from a single batch. All composite samples shall be verified in accordance with 4.4. If any sample fails to comply with any applicable requirements the batch shall be rejected.

4.3.2.2 Sampling at the receiving facility. Ammonium perchlorate shall be sampled using Verification Level II of MIL-STD-1916, based upon the number of shipping containers (e.g., drums, totes, boxes, etc.) in a lot. Each shipping container shall be sampled in three locations in accordance with ASTM E300 for solids. This material shall be placed in a secondary container and mixed to form a composite sample consisting of a minimum of 500 grams for each ammonium perchlorate lot received. There shall be no reduced sampling. Every lot received shall be tested and verified in accordance with 4.3.3. If any sample fails to comply with any applicable requirements the lot shall be rejected.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T05:28Z
Check the source to verify that this is the current version before use.

4.3.3 First article and conformance testing of ammonium perchlorate.				
Category	Examination or test	Verification level	Requirement paragraph	Inspection method
Critical	None defined			
Major				
101	Total moisture	4.3.2	3.2	4.4.1.1
102	Surface moisture	4.3.2	3.2	4.4.1.2
103	Water insolubles	4.3.2	3.2	4.4.2
104	Ash, sulfated	4.3.2	3.2	4.4.3
105	Chlorate	4.3.2	3.2	4.4.4 or 4.4.5
106	Chloride	4.3.2	3.2	4.4.4 or 4.4.6
107	Sodium and potassium	4.3.2	3.2	4.4.7 or 4.4.8
108	Calcium	4.3.2	3.2	4.4.7 or 4.4.9
109	Iron	4.3.2	3.2	4.4.7 or 4.4.10
110	Assay	4.3.2	3.2	4.4.11
111	pH	4.3.2	3.2	4.4.12
112	Granulation	4.3.2	3.2	4.4.13
Minor				
201	Workmanship	4.3.2	3.3	4.4.14

4.4 Methods of inspection. All tests given in this section shall be performed using prescribed procedures for replicate determination given in standard analytical textbooks. Also, unless otherwise specified herein, all chemicals and reagents used in the tests shall be ACS Grade or Reagent Grade. Water in accordance with Type I of ASTM D1193 shall be used as required in all test procedures.

4.4.1 Total moisture and surface moisture. The Karl Fischer method shall be used with any suitable apparatus according to the general procedure described below (see 6.5). The endpoint may be determined either visually or electrometrically by means of either direct or back titration. In all operations, protect the sample against change in moisture content.

a. Standardization of Karl Fischer reagent. Add 100 mL of the pyridine-methanol solution (see 6.5.3) to a clean, dry titration flask containing a stirring bar. Close the flask, start the magnetic stirrer and titrate to an endpoint, using either a direct or back titration. Add 0.10 ± 0.01 grams (g) of sodium tartrate dehydrate (see 6.5.4), keeping the flask open for the minimum time possible, and allow to stir until solution is practically complete. Add small increments of Karl Fischer Reagent (KFR) (see 6.5.1) to assist solution and titrate to an endpoint after solution

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T05:28Z
Check the source to verify that this is the current version before use.

is complete. For back titration, a standard solution of water-in-methanol (WM) should be used (see 6.5.2). Calculate the water equivalence (F) of the KFR as follows:

$$F = \underset{\text{Direct titration}}{\frac{0.1556 \times W}{A}} = \underset{\text{Back titration}}{\frac{0.1556 \times W}{(A \times R) - B}}$$

Where:

A = volume of KFR required in direct titration or added for back titration, mL
B = volume of WM solution used in back titration, mL
R = ratio of volume of standard WM used in back titration (A) to volume of KFR added (B); (R = A/B)
W = weight of sodium tartrate dihydrate, g

Note: the ratio R is determined by adding an accurately measured volume of approximately 10 mL of KFR to a convenient volume of the methanol-pyridine solvent which has just been brought to an endpoint. Titrate the KFR with standard WM solution.

4.4.1.1 Total moisture. Determine total moisture in accordance with either of the following methods.

4.4.1.1.1 Total moisture by volumetric Karl Fischer titration (Grades A, C and D). Place approximately 15 g portions of the sample in dry, stoppered weighing bottles and determine the total weight of sample and bottle to the nearest 10 mg. Add 100 mL of methanol-pyridine solution to the titration vessel and titrate to an end-point with KFR. As soon as the end-point is reached, transfer the contents of the weighing bottle to the titration vessel and immediately reclose the vessel. Do not take time to try to transfer all of the sample, but re-stopper the weighing bottle and again weigh to the nearest 10 mg. Stir the contents of the titration vessel until the sample is dissolved. If necessary, small amounts of KFR may be added to facilitate dissolution. For direct titration, titrate to an end-point with KFR. For back titration, add an excess of KFR and back titrate with WM. Calculate the percent water as follows:

$$\text{Percent water} = \underset{\text{Direct titration}}{\frac{F \times A \times 100}{W}} = \underset{\text{Back titration}}{\frac{F \times ((A \times R) - B) \times 100}{W}}$$

Where:

A = volume of KFR required in direct titration or added for back titration, mL
B = volume of WM solution used in back titration, mL
R = ratio of volume of standard WM used in back titration (A) to volume of KFR added (B)
F = water equivalent of KFR, g/mL (see 4.4.1.a)
W = weight of sodium tartrate dehydrate, g

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T05:28Z
Check the source to verify that this is the current version before use.

Note: the ratio R is determined by adding an accurately measured volume of approximately 10 mL of KFR to a convenient volume of the methanol-pyridine solvent which has just been brought to an endpoint. Titrate the KFR with standard WM solution.

4.4.1.1.2 Total moisture by coulometric Karl Fischer titration (all Grades). Moisture content shall be determined by the Karl Fischer method using coulometric direct-titration technique in accordance with ASTM E1064.

4.4.1.1.2.1 Equipment.

a. Mettler Toledo C30 Coulometric Karl Fischer titrator or equivalent. Equipped with a frit-less or a frit generating electrode.
b. 3 mL or 5 mL syringe, 20 gauge or larger (lower gauge number) with 4 inch cannula or needle.

4.4.1.1.2.2 Reagents and standards.

a. Apura Combi Coulomat frit-less version, 100 mL
b. Hydranal Coulamat AG (fritted version), 100 mL
c. Hydranal Coulamat CG (fritted version), 5 mL
d. Water standard for Karl Fischer analysis (100 ppm / 0.01% or 1% for expected higher water concentration.

4.4.1.1.2.3 Standard analysis.

a. Prior to any sample or standard added, ensure that drift for the coulometric titrator is low (per manufacturer's specification) and stable.
b. Rinse a syringe with approximately 0.5 - 1 mL of water standard. Dispense the solution into waste. Perform rinsing a minimum of two times. Due to hygroscopic nature of the water standard, immediately draw approximately 1 mL of water standard into a syringe. Wipe residual from needle and slightly draw in the solution to prevent spillage while weighing. Weigh the syringe in grams. Record the weight to the nearest 0.0001 g.
c. Initiate the method and allow the solvent to condition to the endpoint.
d. Press the Start Sample key, then dispense the water standard into the titration vessel with the needle or cannula tip slightly below the surface of the solvent. The vessel stopper shall be kept open for the shortest possible period to minimize the introduction of atmospheric moisture into the vessel. The titration will start when the instrument senses the added water. The Start Sample key will not be active until the solvent is conditioned to the endpoint.
e. Reweigh the syringe. Record the weight to the nearest 0.0001 g.
f. Calculate the sample weight in grams. Enter the sample weight into the titrator.
g. When the titration is completed, record the standard moisture percent to the nearest 0.001%. Before continuing, ensure that the standard result is within range specified on the manufacturer’s certificate of analysis for water standard. If the standard is not within range, allow the instrument to equilibrate longer prior to the next water standard injection. If the standard solution is not within range upon second attempt, the water standard shall be replaced.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T05:28Z
Check the source to verify that this is the current version before use.

h. Do not replace the reagent/solvent for after a passing standard has been performed. If solution is changed, another standard analysis shall be performed.

4.4.1.1.2.4 Sample moisture determination preparation by direct sample addition.

a. Initiate the method and allow the solvent to condition to the endpoint. The Start Sample key shall not be active until the solvent is conditioned to the endpoint.

b. Weigh out approximately 0.3 – 0.5 grams of sample into a clean, dry container. Record the weight to the nearest 0.0001 g.

c. Press the Start Sample key, then immediately dispense the sample into the titration vessel. The vessel stopper should be kept open for the shortest possible period to minimize the introduction of atmospheric moisture into the vessel. The titration will start when the instrument senses the added water.

d. Reweigh the container. Record the weight to the nearest 0.0001 g.

e. Calculate the sample weight in grams. Enter the sample weight into the titrator.

f. When the titration is completed, record the sample moisture percent to the nearest 0.001%.

g. If the instrument does not detect any water in the sample, double the sample size and perform again. Doubling sample size a second time shall not be allowed; 4.4.1.1.2.3 shall be re-performed.

h. The moisture determination shall be tested in triplicate. Do not replace the reagent/solvent for each determination (see note below).

i. Record the results as follows: individual values, mean, standard deviation and relative standard deviation (RSD). The relative standard deviation shall not exceed 2.0%. If it does, repeat the sample moisture determination after verifying the vessel seals are secure and the drying agents are active. Note: The coulometric reagent may need to be replaced if any of these conditions applies: the reagent bottle was opened for more than 1 month; the reagent becomes exhausted during the course of multiple individual determinations – this will be evident if the titration is not complete or does not reach equilibrium point; or if the coulometric titrator has been unused for a prolonged period of time, e.g. one month. If so, empty the vessel and add fresh reagents (based on fritted or frit-less version). Allow the solution to condition to the endpoint and resume the analysis. Report the mean for conformance.

Note: Coulometric titrators do not require calibration. However, proper function of the instrument may be checked through the addition of appropriate water standards.

4.4.1.2 Surface or external moisture (all Grades). Proceed as specified in 4.4.1.1.1 except that the solvent shall be the methanol-pyridine mixture saturated with reagent-grade ammonium perchlorate, that has been adequately pre-titrated and titrating as soon as the sample has been transferred to the flask.

4.4.2 Water insolubles. Transfer a specimen of about 50 g (weighed to the nearest 0.1 g) to a 400 mL beaker and add about 250 mL of water. Heat to 75±5˚C and stir until the sample is completely dissolved. Filter the contents of the beaker through a tared, fine porosity, sintered glass crucible and transfer any residue from the beaker to the crucible by washing thoroughly with additional water heated to the same temperature as above. Dry the crucible and contents for

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T05:28Z
Check the source to verify that this is the current version before use.

two hours at 100±5˚C, cool in a desiccator, and weigh. Calculate the water insolubles content of the sample as follows:

$$\text{Percent water insolubles} = 100 \times \frac{A}{B}$$

Where:

A = weight of residue, g
B = weight of sample, g

4.4.3 Sulfated ash. Ignite a clean porcelain, silica or platinum crucible and lid for approximately thirty minutes at 800±25˚C. Cool in a desiccator, weigh to the nearest 0.1 mg, and add approximately 10 g of sample weighed to nearest 0.1 g. Add 3 mL of concentrated sulfuric acid, place the lid on upside down and heat over a Bunsen burner or hot plate until most of the sample is burned off. (Caution: heat the sample in a hood designed specifically for venting perchloric acid fumes and keep a safety shield interposed between operator and sample.) Heat gently since ammonium perchlorate may decompose violently if heated over 300˚C). When volatilization of the sample is essentially complete, transfer the crucible with inverted cover to a muffle furnace, and ignite at 800±25˚C for not less than thirty minutes, and transfer to a desiccator. Replace the crucible cover in an upright position, cool and weigh to the nearest 0.1 mg. Calculate the sulfated ash content as follows:

$$\text{Percent sulfated ash} = 100 \times \frac{C}{D}$$

Where:

C = weight of residue, g
D = weight of sample, g

4.4.4 Chlorate and/or chloride content by anion chromatography. The ammonium perchlorate sample is dissolved in deionized water and the solution analyzed for chloride by suppressed anion chromatography with conductivity detection. The chloride content is determined by comparison to a known standard.

4.4.4.1 Reagents.

a. Deionized water.
b. Sodium chloride, 1000 ppm ACS grade.
c. Sodium chlorate, 1000 ppm ACS grade.
d. 50% NaOH (w/w), high purity (<0.05% sodium carbonate, prepared with deionized water).
e. Chromatography eluent A, 100 mM NaOH. Degas deionized water by vacuum or sparging with high purity helium. Dilute 8.00 g of the 50% NaOH per liter of degassed DI water. The headspace of the container holding the eluent A shall be slightly pressurized by high purity helium during use and storage.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T05:28Z
Check the source to verify that this is the current version before use.

f. Chromatography eluent B. Degas deionized water by vacuum or sparging with high purity helium. The headspace of the container holding the eluent B shall be slightly pressurized by high purity helium during use and storage.

g. Suppressor regenerating solution as recommended by chromatograph manufacturer, typically a 50-100 millimolar solution of sulfuric acid in deionized water.

4.4.4.2 Supplies and equipment.

a. Analytical balance, 200 gram capacity, 0.0001 gram resolution or better.

b. Ion chromatograph with anion suppression and conductivity detector with minimum sensitivity of 100 microsiemens/centimeter full scale, e.g. Dionex/Thermo Fisher Integrion, or equivalent.

c. Ion chromatography column, which together with the proper eluent (4.4.4.1.e-f) can baseline-resolve chloride and completely elute analyte in 30 minutes or less, e.g. Dionex/Thermo Fisher AG-11 guard column with AS-11 analytical column, or equivalent.

d. Volumetric flasks, quantity: 1 x 1 L, 2 x 250 mL, 2 x 100 mL

e. Volumetric pipet, 1 mL, Class A.

f. Prior to starting, all flasks shall be rinsed at least three (3) times with deionized water and allowed to air dry.

4.4.4.3 Preparation of calibration standards.

a. Prepare 1 ppm standard by adding 500 µL of the 1000 ppm sodium chloride standard and 500 µL of the 1000 ppm sodium chlorate standard and diluting to 500 mL with deionized water.

b. Prepare 0.5 ppm standard by adding 250 µL of the 1000 ppm sodium chloride standard and 250 µL of the 1000 ppm sodium chlorate standard and diluting to 500 mL with deionized water.

c. Prepare 0.25 ppm standard by adding 125 µL of the 1000 ppm sodium chloride standard and 125 µL of the 1000 ppm sodium chlorate standard and diluting to 500 mL with deionized water.

NOTE: If sample concentration (see 4.4.4.4.i) is greater than 0.02 weight percent then a fourth calibration standard shall be prepared at or above the measured sample concentration and the analysis re-run.

4.4.4.4 Sample preparation.

a. Weigh 1.0000 ± 0.0050 grams, and record exact weight, of the ammonium perchlorate into a 500 mL volumetric flask. Fill to about half volume with deionized water and swirl to dissolve. Make up to volume with deionized water and mix well.

b. Repeat 4.4.4.4.a to prepare the second replicate.

4.4.4.5 Procedure.

a. Set up the ion chromatograph in accordance with manufacturer instructions. The instrument shall be equilibrated until a steady, linear baseline is obtained. It is recommended to

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T05:28Z
Check the source to verify that this is the current version before use.

inject 1-3 samples of eluent B while the instrument is equilibrating to make sure the chromatograms do not contain any analytes other than water and/or system peaks.

b. Fill up the appropriate instrument vials, one each, with the chloride/chlorate standards (4.4.6.1.3), and the two sample solutions (4.4.4.4.b).

c. All solutions shall be analyzed using a method appropriate for the selected chromatography column (4.4.4.2.c) and system. Allow each run (generally thirty minutes) to completely elute any analyte which may interfere with integration of the subsequent sample.

d. Identify the chloride peak, which elutes during the first half of each run. For the standards, chloride shall be the largest peak in the chromatogram; note its retention time. For the samples, examine the same retention time ± 5%; chloride shall be identified as the largest peak in the same retention time window.

e. Identify the chlorate peak, which elutes during the second half of each run. For the standards, chlorate shall be the second largest peak in the chromatogram; note its retention time. For the samples, examine the same retention time ± 5%; chlorate shall be identified as the largest peak in the same retention time window.

f. Integrate each chromatogram using valley-to-valley type integration making sure the start and endpoints of the chloride peak both coincide with the local baseline minima. Obtain the chloride peak areas for each of the runs.

g. For the standards (4.4.4.3.b), obtain a linear calibration of peak area versus ammonium chloride concentration using a spreadsheet or software included with the instrument. Perform the same for ammonium chlorate.

h. From the calibration line, calculate the ammonium chloride concentration (in milligrams per liter) corresponding to the chloride peak area for each of the sample replicates (4.4.4.4.b). Perform the same for ammonium chlorate concentration.

i. Calculate percent chloride and percent chlorate as follows:

$$\% \, NH_4Cl = 100 \times \frac{S_{NH_4Cl} \times 0.5}{W_S \times 1000}$$

$$\% \, NH_4ClO_3 = 100 \times \frac{S_{NH_4ClO_3} \times 0.5}{W_S \times 1000}$$

Where: S_{NH4Cl} = concentration of ammonium chloride, mg/L (4.4.4.5.h)
$S_{NH4ClO3}$ = concentration of ammonium chlorate, mg/L (4.4.4.5.h)
W_S = sample weight, g (4.4.4.4.b)

j. Calculate and report the arithmetic mean of the two % NH_4Cl and % NH_4ClO_3 values found in 4.4.4.5.h for conformance.

4.4.5 Chlorate content by absorption of light.

4.4.5.1 Chlorate content by color comparison.

4.4.5.1.1 Orthotolidine dihydrochloride (o-tolidine) reagent. Dissolve 1.35 g of o-tolidine in 500 mL of water. Add this solution, with constant stirring, to a mixture of 350 mL of water and 150 mL of concentrated hydrochloric acid. This reagent should be stored at room

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T05:28Z
Check the source to verify that this is the current version before use.

temperature in amber bottles or in the dark, protected from sunlight and contact with rubber. The reagent shall be discarded after six months.

4.4.5.1.2 Standard solution. Dissolve 0.212 g of sodium chlorate in water and dilute to one liter.

4.4.5.1.3 Procedure. Transfer 2.00 g of the sample and 1.8, 1.9, 2.0, and 2.1 mL aliquots of the standard solution into five separate, glass-stoppered Erlenmeyer flasks. Add 50 mL of water and 20 mL of hydrochloric acid to each. Maintain at 40˚C for thirty minutes, cool to room temperature, and transfer to 100 mL Nessler tubes containing 5 mL of o-tolidine reagent. Dilute to 100 mL with water, mix, and allow the color to develop for not less than five minutes, preferably in the dark. Choose the standard solution that is comparable to the sample solution. Calculate the ammonium chlorate content based on the matching standard as follows:

$$\text{Percent } NH_4ClO_3 = 0.01 \times A$$

Where:

A = volume of aliquot in comparable standard, mL

Note: Permanent standards may be prepared from copper sulfate and potassium dichromate (see 6.8).

4.4.5.2 Chlorate content by UV/VIS spectrometry (alternate).

4.4.5.2.1 Reagents.

a. 1:3 hydrochloric acid (concentrated, 37%) to water
b. Standard 0.06 g/L potassium chlorate solution
c. O-tolidine solution (same as 4.4.5.1.1)

4.4.5.2.2 Apparatus. UV/VIS spectrometer with a range of 190 nm to 1100 nm, e.g. Perkin Elmer Lambda Bio or equivalent.

4.4.5.2.3 Preparation of test solutions.

a. Blank solution: Add 80 mL of 1:3 hydrochloric acid solution to a clean 100 mL volumetric flask and heat to 40˚C for thirty minutes.

b. Standard solutions: In a clean 100 mL volumetric flask, add 1.0 mL of the standard 0.06 g/L potassium chlorate solution and 80 mL of the 1:3 hydrochloric acid solution. To another clean 100 mL volumetric flask, add 2.0 mL of the potassium chlorate solution and 80 mL of 1:3 hydrochloric solution. Heat both flasks to 40˚C for thirty minutes. The concentrations of the standard solution equate to 0.50 and 1.0 micrograms of ammonium chlorate per milliliter, respectively, or concentrations of 0.005% and 0.01% ammonium chlorate when applied to a 1.000 g sample.

c. Preparation of sample solutions: Weigh 0.5 g of sample, accurate to 0.01 mg, and place in a clean 100 mL volumetric flask. Add 80 mL of 1:3 hydrochloric acid solution and heat at 40˚C for thirty minutes. Develop color in blank, standards, and sample solutions by cooling to

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T05:28Z
Check the source to verify that this is the current version before use.

room temperature, adding 1 mL of o-tolidine solution and diluting to mark volume with water. After mixing, the solutions are allowed to stand for five minutes to complete color development.

4.4.5.2.4 Procedure.

a. Set wavelength at 490 nm and introduce the blank solution. Turn selector to concentration mode and set scale readout to zero. Introduce the high concentration standard solution, the 0.01% NH_4ClO_3 equivalent, and, by use of the concentration adjustment control, set the readout to 0.01.

b. For linearity verification, introduce the 0.005% standard solution, and the readout should show 0.005. Readouts of less than 0.0045 or greater than 0.0054 indicate linearity has not been achieved and the solutions should be reintroduced in order, with appropriate control adjustment. If the verification is not successful, prepare a new blank and standard solution. Introduce sample solution, prepared in the exact manner described, and read the sample concentration in percent ammonium chlorate (NH_4ClO_3) directly from the instrument display readout. Additional instructions may be obtained from the instrument operator's manual.

4.4.6 Chloride. Dissolve 20 g of the specimen (weighed to the nearest 0.05 g) in 150 mL of water in an Erlenmeyer flask and add 3 to 4 mL of nitric acid (ACS grade). Add standardized 0.05 Normal (N) silver nitrate solution (see 6.6.1) from a buret until the precipitate coagulates, mix thoroughly, and allow to stand without mixing. Allow precipitate to settle to the bottom of the flask. Add several drops of silver nitrate solution from the buret to the clear supernatant liquid. If precipitation is observed, mix and allow to settle again. Continue adding several drops of silver nitrate until no precipitation is observed. Once precipitation is complete (no marked cloudiness occurs), add an additional 5 mL accurately of the silver nitrate solution. With mixing, add 1 to 2 mL of nitrobenzene and 3 mL of ferric ammonium sulfate indicator. Mix for five minutes and allow precipitate to settle. The supernatant liquid shall not be milky – if it is then prepare fresh silver nitrate and ferric ammonium sulfate indicator solutions and repeat procedure. Begin mixing again and titrate with 0.05N potassium thiocyanate solution (see 6.6.3) to a permanent red-brown color. Calculate the ammonium chloride content as follows:

$$\text{Percent ammonium chloride} = \frac{5.35 \times ((a \times A) - (b \times B))}{W}$$

Where:

A = volume of silver nitrate solution, mL
B = volume of potassium thiocyanate solution, mL
a = normality of silver nitrate solution
b = normality of potassium thiocyanate solution
W = weight of sample, g

Alternatively, chloride may be determined by direct potentiometric titration with combined silver ring electrode/reference electrode. If this method is selected, dissolve 20 g of the specimen (weighed to the nearest 0.05 g) in 150 mL of water in an Erlenmeyer flask and add 3 to 4 mL of nitric acid (ACS grade). Titrate to endpoint potentiometrically using standardized 0.05N silver nitrate solution (see 6.6.1). Calculate the ammonium chloride content as follows:

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T05:28Z
Check the source to verify that this is the current version before use.

$$\text{Percent ammonium chloride} = \frac{5.35 \times (a \times A)}{W}$$

Where:

A = volume of silver nitrate solution, mL
a = normality of silver nitrate solution
W = weight of sample, g

4.4.7 Trace metal impurities by Inductively Coupled Plasma – Optical Emission Spectroscopy (ICP-OES) (sodium, potassium, calcium, and iron). Class A glassware shall be used throughout this method.

4.4.7.1 Reagents. It is advisable to use commercially available stock solutions with certified concentrations traceable to primary standards (National Institute of Standards and Technology or international measurement standards). Stock solutions shall be replaced after one year or in accordance with manufacturer recommendations. Alternately, stock solutions may be prepared in-house from high purity metals or their salts; these materials shall be Reagent grade or better. The maximum recommended shelf life is one year from date of initial preparation when kept tightly sealed at 5-25°C. Common commercially available stock concentrations are as follows:

a. Certified Na stock solution, 100 ± 2 mg/L in 2-5% (v/v) HNO_3
b. Certified Ca stock solution, 100 ± 2 mg/L in 2-5% (v/v) HNO_3
c. Certified Fe stock solution, 100 ± 2 mg/L in 2-5% (v/v) HNO_3
d. Certified K stock solution 100 ± 2 mg/L in 2-5% (v/v) HNO_3

Additional necessary reagents are:

e. De-ionized water.
f. Nitric acid (HNO_3), concentrated, 67-70%, trace metal grade.
g. Nitric acid (HNO_3), 3%, made by diluting 22 ± 2 mL of concentrated nitric acid (4.4.7.1.f) to 500 mL with deionized water; this is the minimum recommended volume of 3% nitric acid required for analysis.
h. Compressed argon, as per ICP-OES manufacturer recommendations.

4.4.7.2 Preparation of reference (standard) solutions.

4.4.7.2.1 Reference solutions for iron. Prepare 0.2 mg/L, 1 mg/L, and 2 mg/L standards. If using the stock concentration listed in 4.4.7.1.c then use the following steps.

a. Pipette 0.2 mL of the certified stock solution into a 100 mL volumetric flask. Fill to the mark with 3% nitric acid and stopper the flask. Invert several times to mix. This is the 1 mg/L standard.
b. Pipette 1 mL of the certified stock solution into a 100 mL volumetric flask. Fill to the mark with 3% nitric acid and stopper the flask. Invert several times to mix. This is the 5 mg/L standard.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T05:28Z
Check the source to verify that this is the current version before use.

c. Pipette 2 mL of each certified stock solution into a 100 mL volumetric flask. Fill to the mark with 3% nitric acid and stopper the flask. Invert several times to mix. This is the 10 mg/L standard.

4.4.2.2.2 Reference solutions for sodium, calcium, and potassium. Prepare 1 mg/L, 5 mg/L, and 10 mg/L standards. If using the stock concentrations listed in 4.4.7.1.a, b, and d then use the following steps.

a. Pipette 1 mL of each certified stock solution into a 100 mL volumetric flask. Fill to the mark with 3% nitric acid and stopper the flask. Invert several times to mix. This is the 1 mg/L standard.
b. Pipette 5 mL of each certified stock solution into a 100 mL volumetric flask. Fill to the mark with 3% nitric acid and stopper the flask. Invert several times to mix. This is the 5 mg/L standard.
c. Pipette 10 mL of each certified stock solution into a 100 mL volumetric flask. Fill to the mark with 3% nitric acid and stopper the flask. Invert several times to mix. This is the 10 mg/L standard.
d. Prepare a fourth flask of 3% nitric acid and retain as the blank solution.

4.4.7.3 Sample preparation.

a. Weigh 1000 mg ± 50 mg of sample to the nearest mg and transfer into a 100 mL volumetric flask.
b. Add approximately 90 mL of the 3% nitric acid to the flask and place the stopper on the flask.
c. Allow the sample to completely dissolve. Agitation by means of rotary shaker or ultrasonic bath may be used if necessary.
d. Fill to the mark with 3% nitric acid and invert several times to mix.

4.4.7.4 Instrument preparation. Prepare the instrument according to manufacturer instructions. It is recommended to perform all measurements in radial mode. Instrument response for each analyte shall be such that the response of the 1 mg/L standard is greater than ten times that of the blank, and the response of the 10 mg/L solution does not saturate the detector. Ensure a steady baseline with no drift; measure the response of the 5 mg/L standard at 5 minute intervals until successive measurements differ by no more than 5%. Allow for adequate flush time of the system between standard and sample solutions to ensure there is no carryover. Calibrate the system for all four analytes using the standard solutions and blank. Analyze the samples in the same run, recording the results in mg/L. Recommended wavelengths used for each element are in Table VII. Other wavelengths can be used however interferences need to be ruled out.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T05:28Z
Check the source to verify that this is the current version before use.

TABLE VII. Recommended wavelengths for ICP-OES element analysis.

Element	Wavelength (nm)
Fe	259.940
Ca	393.366
Na	589.592
K	766.491

4.4.7.5 Analysis.

a. For each element generate a four-point calibration line utilizing the prepared standards and the blank. Force the line through zero concentration. The line shall have a best fit value (R^2) of not less than 0.95.

b. Aspirate each sample for the time required by the instrument to achieve a stable response. Measure the response and calculate the concentration of each element from its corresponding calibration line, recording the results in mg/L.

c. If the calculated concentration is higher than 10 mg/L, perform a 10-fold dilution of the sample (e.g. 5 mL of original sample solution diluted to 50 mL with 3% nitric acid) and repeat previous step using the 10-fold-diluted sample. Multiply the result by 10 to determine the original concentration in mg/L.

d. Calculate the percent compound for direct comparison to the chemical requirements in Table I as follows. Multiply the measured concentration in mg/L for each element by the corresponding conversion factor in Table VIII Multiply the measured concentration in mg/L for of iron by the conversion factor 0.0143 (to convert to Fe_2O_3) and report for conformance. Add the measured concentrations in mg/L for sodium and potassium and report for conformance.

Table VIII. Conversion factors.

Element	Chemical requirement	Factor to convert mg/L to sample wt.%
Ca	Calcium (as Ca)	0.0100 / sample weight
Fe	Iron	0.0100 / sample weight
Fe	Iron (as Fe_2O_3)	0.0143 / sample weight
Na	Sodium (as Na)	0.0100 / sample weight
K	Potassium	0.0100 / sample weight

4.4.8 Sodium (Na) and potassium (K) (alternate (see 4.4.7)).

4.4.8.1 Preparation of reagents. Weigh approximately 1.9 g of potassium chloride and approximately 2.5 g of sodium chloride, each accurately recorded to 0.0001 g, and quantitatively transfer into separate 1-L volumetric flasks containing approximately 250 mL of distilled water. Swirl to dissolve the sample and dilute to volume with water. Each solution contains approximately 1 mg/mL of cation. From these stock solutions transfer aliquots to other volumetric flasks and dilute with water so that at least four standard solutions are prepared in the range 0.005 to 0.03 mg/mL for K ion and 0.001 to 0.005 mg/mL for Na ion.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T05:28Z
Check the source to verify that this is the current version before use.

4.4.8.2 Preparation of standard curves. Determine the net emission in terms of absorbance or percent transmittance of the standard solutions using a wavelength of 766.5 nm for K ion and 589 nm for Na ion. The minimum practical slit width should be used. Plot absorbance vs. concentration on linear coordinates or percent transmittance vs. concentration on semi-log coordinates. Fit a straight line to the points.

4.4.8.3 Analysis of sample. Quantitatively transfer 4.0 g of ammonium perchlorate, weighed to the nearest mg, into a 100 mL volumetric flask. Dissolve in distilled water and dilute to volume. Determine the absorbance or percent transmittance of each standard solution, a distilled water blank solution, and the sample solution, by flame photometry using the manufacturer's manual of instructions for the spectrophotometer. Read the concentration of the sample directly from the standard curve. Calculate the potassium and sodium content as follows:

$$\text{Potassium and sodium, wt. percent} = \frac{10 \times (R_K + R_{Na})}{W}$$

Where:

R_K = potassium concentration as read from standard curve, mg/mL
R_{Na} = sodium concentration as read from standard curve, mg/mL
W = weight of sample, g/100 mL

4.4.9 Tricalcium phosphate content (alternate (see 4.4.7)). Determine tricalcium phosphate content in accordance with either of the following procedures.

4.4.9.1 Tricalcium phosphate content.

4.4.9.1.1 Preparation of reagents.

a. Ammonium vanadate solution: Add 2.350 g of ammonium metavanadate to approximately 400 mL of warm distilled water in a 1 L volumetric flask. Slowly and cautiously add 14 mL of 70-72% perchloric acid, mix to effect solution, cool to room temperature, and dilute to volume with water.

b. Ammonium molybdate solution: Dissolve 100 g of molybdic acid in a mixture of 300 mL of distilled water and 80 mL of ammonium hydroxide. Filter the solution and boil the filtrate for 20 minutes. Cool, transfer to a 1 L volumetric flask, and dilute to volume with distilled water.

c. Perchloric acid: 70-72% ACS grade.

4.4.9.1.2 Apparatus. Bausch and Lomb "Spectronic" 710 or equivalent.

4.4.9.1.3 Preparation of calibration curve. Quantitatively transfer 5.00 g of alpha-tricalcium phosphate (α-TCP, ≥98%) to a 1-L volumetric flask, add 500 mL of water and 20 mL of perchloric acid. Mix to dissolve and dilute to volume with distilled water. Transfer 1, 2, 3, 4, and 5 mL aliquots to separate 100 mL volumetric flasks. Add to each flask 25 mL of water and 17 mL of perchloric acid. Cool to room temperature and add 10 mL of ammonium vanadate solution. Dilute to 75 mL with water, mix and cool to room temperature. Add 10 mL of

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T05:28Z
Check the source to verify that this is the current version before use.

ammonium molybdate solution, dilute to volume with water and let stand at room temperature for at least thirty minutes. Prepare a blank solution containing all the reagents in the specified amounts but without the TCP. Using the blank solution in the reference cell, determine the absorbance of each of the five standard solutions at 470 nm. Prepare a calibration curve by plotting absorbance vs. concentration, expressed as g TCP / 100 mL of solution, and draw the calibration curve by fitting the best straight line to the points.

4.4.9.1.4 Procedure. Quantitatively transfer 2.50 g of sample to a 100 mL volumetric flask and dissolve in about 50 mL of distilled water. Add, slowly and cautiously, 20 mL of perchloric acid and, after thorough mixing, cool to room temperature. Add 10 mL of ammonium vanadate solution, dilute to 75 mL with water, mix and cool to room temperature. Add 10 mL of ammonium molybdate solution, dilute to volume with water, mix well and allow to stand at room temperature for not less than thirty minutes. Determine the absorbance of the sample solution at 470 nm, using the blank solution in the reference cell. Determine the weight of the TCP by referring to the calibration curve and calculate the amount in the sample by substituting in the following equation:

$$\text{TCP, percent} = \frac{100 \times A}{W}$$

Where:

A = weight of TCP in sample, g
W = weight of sample, g

4.4.9.2 Tricalcium phosphate (alternate).

4.4.9.2.1 Reagents.

a. Ammonium molybdate solution (10% aqueous perchloric acid)
b. Ammonium phosphate (dibasic) 0.213 g/250 mL
c. Ammonium vanadate solution, 0.25% aqueous

4.4.9.2.2 Apparatus. Bausch and Lomb “Spectronic” 710 or equivalent.

4.4.9.2.3 Preparation of test solutions.

a. Blank solution: In a clean 100 mL volumetric flask, place 25 mL of distilled water and 17 mL of concentrated perchloric acid. Cool to room temperature and add 10 mL of ammonium vanadate solution (0.25%). Dilute to 75 mL with water and add 10 mL of ammonium molybdate solution (10%). Dilute to mark volume with water, mix well, and allow to stand for thirty minutes for maximum color development.

b. Standard solutions: In two clean 100 mL volumetric flasks, place respectively, 1.0 and 2.0 mL of ammonium phosphate standard solution. In order, treat each flask as the blank was treated. The concentrations of the standard solutions equate to 0.10 and 0.20% TCP, respectively.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T05:28Z
Check the source to verify that this is the current version before use.

c. Preparation of sample solutions: Weigh a 25 g sample to the nearest 0.1 mg and transfer it to a 250 mL volumetric flask. Add about 50 mL of water and mix well. Carefully add 25 mL of concentrated perchloric acid to the flask, swirling constantly, and cool to room temperature. Dilute to volume and mix well. Pipet a 10 mL aliquot into a 100 mL volumetric flask. Add 20 mL of water. Add 16 mL of concentrated perchloric acid. Add 10 mL of ammonium vanadate (0.25%), dilute to about 75 mL with water, swirling constantly, and cool to room temperature. Add 10 mL of ammonium molybdate (10%), dilute to volume, and mix well. Allow the solution to stand for at least thirty minutes at room temperature.

4.4.9.2.4 Procedure.

a. Set wavelength at 470 nm and introduce the blank solution. Turn selector to concentration mode and set scale readout to zero. Introduce the 0.20% TCP standard solution and, by use of the concentration adjustment control, set the readout to 0.20.

b. For linearity verification, introduce the 0.10% TCP standard solution. The display readout should show 0.10. If linearity is not achieved, the solutions should be introduced in order, with appropriate control adjustment and, if this fails, new blank and standard solutions prepared. Introduce sample solutions prepared in the exact manner described, and read the sample concentration, in percent TCP, directly from the instrument display readout. Additional instructions may be obtained from the instrument operator's manual.

4.4.10 Iron content. Determine iron content in accordance with the following method.

4.4.10.1 Preparation of standard iron solution. Dissolve 0.864 g of ferric alum, $FeNH_4(SO_4)_2 \cdot 12\ H_2O$ in 100 mL of sulfuric acid (H_2SO_4) solution prepared by cautious addition of 5 mL of H_2SO_4 to 95 mL of water. When solution is complete and at room temperature (20-25˚C), dilute to 1 L in a volumetric flask.

4.4.10.2 Procedure. Dissolve 7.00 g of the specimen in 75 mL of distilled water. Add 2.0 mL of concentrated H_2SO_4 and transfer to a 100 mL Nessler tube. Add 2 drops of 1 percent potassium permanganate solution and 10 mL of 2 percent ammonium thiocyanate. Dilute to 100 mL and mix well. Into a second Nessler tube containing the same amounts of water and reagents as above, pipette 1.75 mL of the standard iron solution. Dilute and mix well. If the unknown is no darker than the standard, the specimen meets the requirement for iron.

4.4.11 Assay. Determine assay in accordance with the following methods for the appropriate Grade.

4.4.11.1 Assay. This method measures assay by dissolution and titration with sodium methylate (a.k.a. sodium methoxide) using thymol blue as indicator.

4.4.11.1.1 Apparatus and reagents.

a. Methyl ethyl ketone
b. N, N-dimethyl formamide

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T05:28Z
Check the source to verify that this is the current version before use.

c. Thymol blue indicator, 0.2% in methanol. Dissolve 0.2 g of thymol blue in 100 g methanol.

d. Sodium methylate titrant. In a 1 liter flask add 30 mL sodium methylate to 300 mL methanol. Bring to 1 liter with benzene.

e. Benzoic acid, reagent grade

4.4.11.1.2 Standardization of sodium methylate titrant.

a. Weigh between 0.35 and 0.50 g benzoic acid and record.

b. Dissolve in a mixture of 30 mL methanol and 70 mL benzene. Quickly stopper flask

c. Add 10 drops of 0.2% thymol blue indicator, replace the stopper with the stopper containing the titration tip, and titrate to a blue endpoint with the standard sodium methylate titrant.

d. Run a blank using 30:70 methanol:benzene without benzoic acid as described in c.

e. Calculate the normality of the sodium methylate titrant as:

$$\text{Normality} = \frac{\text{grams benzoic acid} \times 8.1885}{\text{mL titrant for sample} - \text{mL titrant for blank}}$$

4.4.11.1.3 Sample analysis.

a. Weigh and record approximately 0.700 g sample.

b. Dissolve sample in a 1:1 mix of 50 mL methyl ethyl ketone with 50 mL N, N-Dimethyl formamide in a 250 mL Erlenmeyer flask. Quickly stopper flask.

c. Add 10 drops of 0.2% thymol blue indicator, replace the stopper with the stopper containing the titration tip, and titrate to a blue endpoint with the standard sodium methylate titrant.

d. Carry a titration blank throughout the process (steps b-c) using all reagents without the sample.

e. Calculate the ammonium perchlorate assay as:

$$\text{Assay } NH_4ClO_4 = \frac{11.75 \times (A - B) \times N}{W} - ((2.20 \times C) + (1.16 \times D))$$

Where:
A = sample titration, mL
B = blank titration, mL
N = normality of standard sodium methylate (see 4.4.12.1.2)
C = ammonium chloride in sample, wt. % (from 4.4.6)
D = ammonium chlorate in sample, wt. % (from 4.4.5)
W = sample weight, g
2.20 = conversion factor of ammonium chloride to ammonium perchlorate
1.16 = conversion factor of ammonium chlorate to ammonium perchlorate
11.75 = (117.5 g/equivalent of NH_4ClO_4) x (1 L/1000 mL) x 100%

4.4.11.2 Assay (alternate). Transfer an accurately weighed portion of approximately 0.5 g of the specimen to a platinum crucible containing a known weight of approximately 5 g of

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T05:28Z
Check the source to verify that this is the current version before use.

sodium carbonate and mix carefully. Overlay the mix with a known weight of sodium carbonate (approximately 1 g) so that a correction can be made for any chloride in the sodium carbonate. Fuse the mixture over a low flame for approximately thirty minutes and then gradually increase the flame to full heat. Dissolve the fusion in a covered beaker containing 50 mL of distilled water and add 50 mL of 1 nitric acid : 1 water. Quantitatively transfer this solution to a 500 mL Erlenmeyer flask and dilute to about 150 mL with water. Add exactly 50 mL of 0.1N silver nitrate solution (see 6.6.1), mix well, add 2 mL of nitrobenzene, and mix again. The supernatant liquid shall not be milky. Add 2 to 3 mL of ferric alum indicator solution (see 6.6.3) and back titrate with 0.1N potassium thiocyanate solution (see 6.6.2) to the first detectable color change which persists for at least thirty seconds.

Alternatively, perchlorate may be determined by a direct electrometric titration. If this method is selected, fuse the sample with sodium carbonate and dissolve in nitric acid as directed above. Then titrate with 0.1N silver nitrate solution to a permanent endpoint using a silver indicator electrode and a glass reference electrode.

With either method, conduct a blank determination to correct for titratable impurities. Calculate the ammonium perchlorate as follows:

$$\text{Percent ammonium perchlorate} = \underset{\text{Back titration}}{\frac{11.75 \times N \times (A - B)}{W}} - ((2.20 \times C) + (1.16 \times D))$$

$$= \underset{\text{Direct titration}}{\frac{11.75 \times N' \times (A' - B')}{W}} - ((2.20 \times C) + (1.16 \times D))$$

Where:

N = normality of potassium thiocyanate
N’ = normality of silver nitrate
A = volume of potassium thiocyanate required for blank titration, mL
A’ = volume of silver nitrate required for sample titration, mL
B = volume of potassium thiocyanate required for sample titration, mL
B’ = volume of silver nitrate required for blank titration, mL
C = percent ammonium chloride (see 4.4.6)
D = percent ammonium chlorate (see 4.4.5)
W = weight of sample, g

4.4.11.3 Assay (alternate).

a. Accurately weigh 5.0 g of dried sample and dissolve in 40 mL of water in a glass-stoppered flask.
b. Mix 20 mL of formaldehyde with 20 mL of water and add to flask.
c. Add 3 drops of phenolphthalein and neutralize with 1N NaOH to endpoint.
d. Mix, seal flask to avoid exposure to air, allow to stand thirty minutes, then add additional 1N NaOH to a pink color which persists for five minutes.
e. Calculation:

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T05:28Z
Check the source to verify that this is the current version before use.

$$\text{Percent ammonium perchlorate} = \frac{11.75 \times A \times N}{W} - ((2.20 \times C) + (1.16 \times D))$$

Where:

A = total volume of NaOH used for titration, mL
N = normality of NaOH
C = ammonium chloride in sample, wt. % (from 4.4.6)
D = ammonium chlorate in sample, wt. % (from 4.4.5)
W = weight of sample, g

4.4.12 pH. Add 50 mL of freshly boiled water to 20.0 g of specimen. Stir and cool to room temperature (20-25˚C). Determine the pH of this solution in accordance with ASTM E70.

4.4.13 Granulation. The granulation shall be determined in accordance with MIL-STD-286, Granulation Method 506.1, except the gyration rate shall be 268 to 315 gyrations per minute.

4.4.14 Workmanship. Take approximately 300 grams of the material and spread the composition out on a clean white surface. The sample shall be examined by personnel with normal vision. The sample shall be uniform in quality and free from odor, visible impurities, and foreign material such as dirt, wooden particles or any other foreign matter. Failure to meet the workmanship inspection or during following production processes shall result in batch/lot rejection.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point' packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Grade A ammonium perchlorate is intended for use in rocket propellants, in tracer ammunition, and in flame throwers. Grades B and C ammonium perchlorate is intended for use in rocket propellants. Grade D ammonium perchlorate is intended for use in the M864 propellant and has historically only been procured in Class 8 granulation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T05:28Z
Check the source to verify that this is the current version before use.

6.2 Acquisition requirements. Acquisition documents should specify the following:
a. Title, number, and date of this specification.
b. Grade and Class required.
c. Whether a first article sample is required (see 3.2).
d. Packaging requirements (see 5.1).
e. Whether acceptance and description sheets are required (see 6.2.1)
f. Iron content (see Table I).

6.2.1 Data requirements. When this specification is used in a procurement contract which incorporates DD Form 1423, acceptance and description sheets prepared in accordance with MIL-STD-1171, or as directed in the contract should be supplied by the contractor.

6.3 Submission of inspection equipment designs and verification methods for approval. Equipment designs, analysis procedures, and accuracy and precision data of the procedures should be submitted, as required, to the Technical Agency (see 6.4)

6.4 Technical agency. Picatinny Arsenal is the technical agency that is responsible for the preparation and revision of the specification. All correspondence to the Technical Agency should be submitted through the contracting officer to: DEVCOM – AC, ATTN: FCDD-ACE-QEE, Picatinny Arsenal, NJ 07806-5000.

6.5 Total moisture determination. The solutions and standards required for the Karl Fischer tests may be prepared as follows.

6.5.1 Karl Fischer Reagent. Dilute stabilized KFR (water equivalence 5.7 mg/mL) with absolute methyl alcohol (0.1% water maximum) or stabilized KFR diluent, so that the water equivalence is 1.4 – 2.3 mg/mL.

6.5.2 Water-in-methanol solution. Add sufficient water to absolute methanol so that the final concentration is approximately 1 mg/mL

6.5.3 Pyridine-ethanol solution. Mix three volumes of anhydrous pyridine with one volume of anhydrous methanol. If necessary, the solvents may be made anhydrous (0.01% H_2O, maximum) by the methods given below.

6.5.3.1 Pyridine. Add one volume of benzene to nineteen volumes of pyridine and distill until 5% of distillate has been collected. Discard the distillate and use the dry residual 95%.

6.5.3.2 Methanol. Distill the methanol over magnesium.

6.5.4 Sodium tartrate dehydrate standard. The water content of the standard may be verified by drying a 2 to 3 g sample in a wide-mouth weighing bottle at 155±5˚C for four hours. The loss in weight should not be less than 15.61% nor greater than 15.71%.

6.6 Standard and indicator solutions. The solutions required for the determination of chloride content (see 4.4.6) and assay (see 4.4.11.2) may be prepared as follows.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T05:28Z
Check the source to verify that this is the current version before use.

6.6.1 Silver nitrate solutions. The 0.1N solution (nominal) may be prepared as described in MIL-STD-1234, Method 607.1. An aliquot of this solution may be diluted with an equal volume of water to give a nominal 0.05N solution.

6.6.2 Potassium thiocyanate solutions. The 0.05N solution (nominal) may be prepared and standardized as described in MIL-STD-1234, Method 608.1. An aliquot of this solution may be diluted with an equal volume of water to give a nominal 0.05N solution.

6.6.3 Ferric alum indicator. This solution may be prepared from $FeNH_4(SO_4) \cdot 12\ H_2O$ as described in MIL-STD-286, Method 705.1.

6.7 Approval of equivalent test methods. Prior approval of the Contracting Officer is required for use of equivalent test methods. A description of the proposed method should be submitted through the Contracting Officer to the Technical Agency (see 6.4). This description should include but not be limited to the procedures used, the accuracy and precision of the method, test data to demonstrate the accuracy and precision and drawings of any special equipment required.

6.8 Chlorate test. For further information on the o-tolidine reagent and permanent standards for chlorate test, refer to the Standard Methods for the Examination of Water and Wastewater, 11th edition, pages 85-91, published by the American Public Health Association (see https://www.standardmethods.org/).

6.9 History of changes. ECP R23Q2007 created revision C. This ECP adopted the Grade D requirements and test methods from Interim Amendment 2, and added new verification methods for chlorate, chloride, sodium, potassium, calcium, iron, and assay determinations.

6.10 Subject term (key word) listing.

Base bleed
Base burn
Propellant
M864
XM1128

6.11 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T05:28Z
Check the source to verify that this is the current version before use.

CONCLUDING MATERIAL

Custodian:
Army – AR
Navy – OS
Air Force – 19

Preparing activity:
Army – AR

(Project 6810-2023-002)

Review Activities:
Army – EA
Navy – AS
Air Force – 68

Other:
DLA - GS

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T05:28Z
Check the source to verify that this is the current version before use.